

{3} Current Electricity

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- * ELECTRIC CURRENT
- * ELECTRIC CURRENTS IN CONDUCTORS
- * OHM' S LAW
- * DRIFT OF ELECTRONS AND THE ORIGIN OF RESISTIVITY
- * LIMITATIONS OF OHM 'S LAW

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* DRIFT OF ELECTRONS AND THE ORIGIN OF RESISTIVITY

* LIMITATIONS OF OHM 'S LAW

* TEMPERATURE DEPENDENCE OF RESISTIVITY

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- * In the present chapter, we shall study some of the basic laws concerning **steady electric currents**.

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- * The direction of current is taken as the direction of motion +ve charges or opposite to the motion of -ve charge.

Ex 1 : In cosmic rays 0.15 protons/cm²-s are entering the earth's atmosphere. If the radius of the earth is 6400 km, what is the current received by the earth in the form of cosmic rays?

Ex 2: In neon gas discharge tube 2.9×10^{18} Ne^+ ions move to the right through a cross-section of the tube each second, while 1.2×10^{18} electrons move to the left in this time. What is the net electric current in the tube?

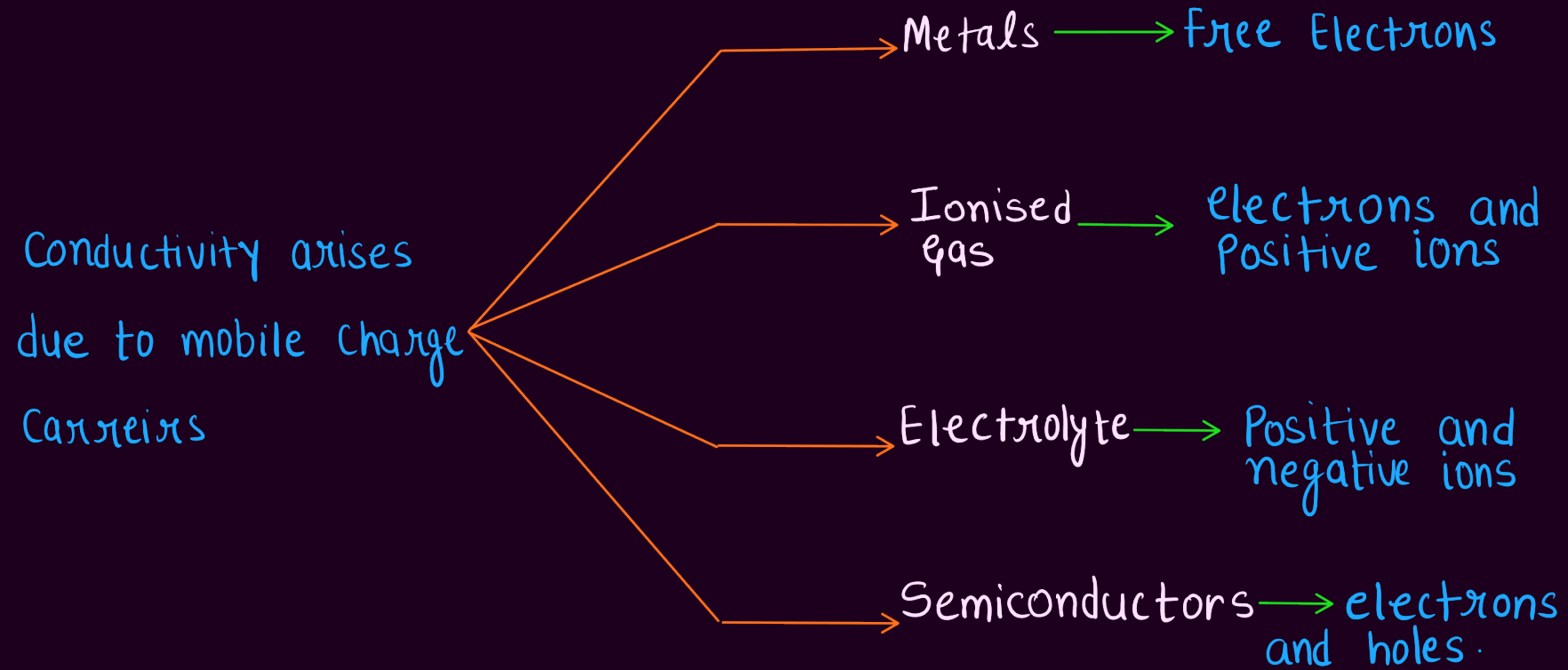
Ex 3 : In the Bohr's model of hydrogen atom, the electron is assumed to rotate in a circular orbit of radius 5×10^{-11} m, at a speed of 2.2×10^6 m/s. What is the current associated with electron motion?

Ex 4 : The current in a wire varies with time according to the relation $I = 2 \text{ A} + (0.6 \text{ A/s}^2) t^2$.

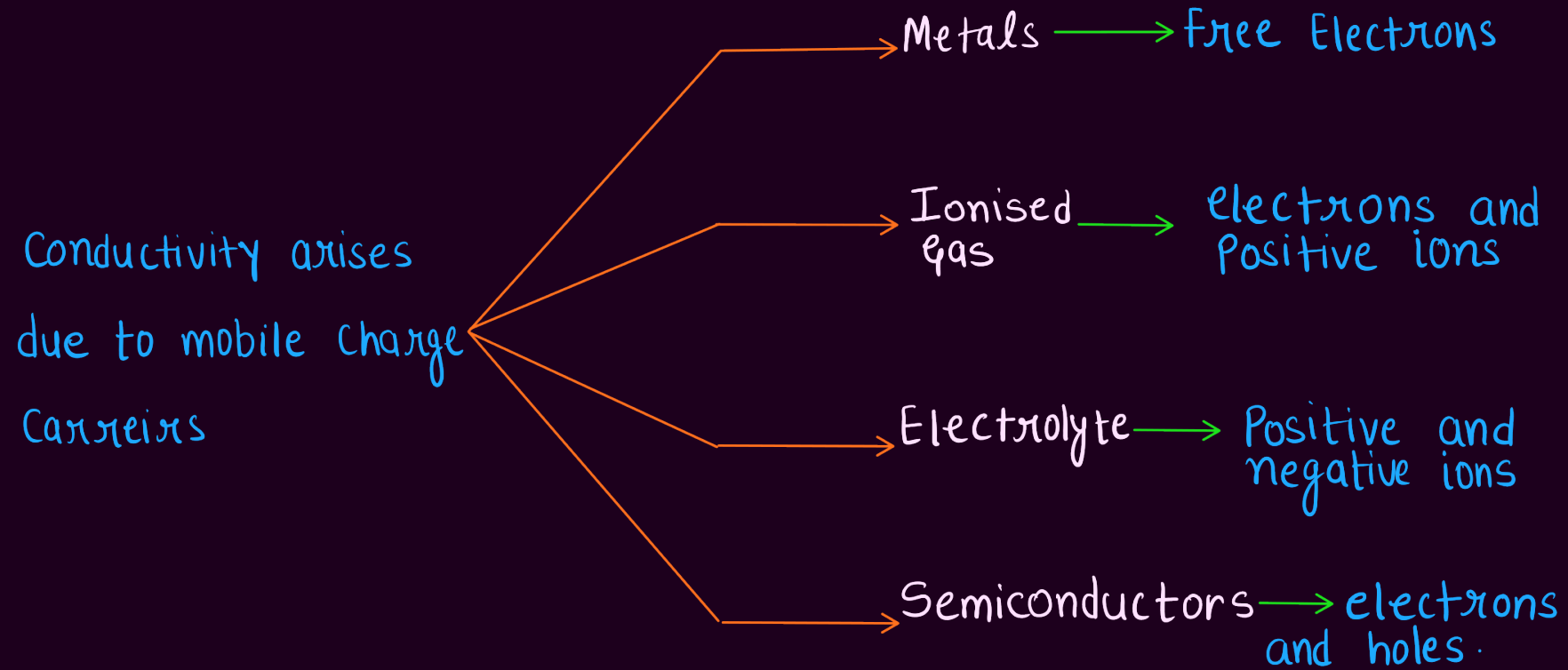
- (a) How many coulomb of charge pass a cross-section of the wire in the time interval between $t = 0$ and $t = 10 \text{ s}$?
- (b) What constant current would transport the same charge in same charge in the same time interval?

ELECTRIC CURRENTS IN CONDUCTORS

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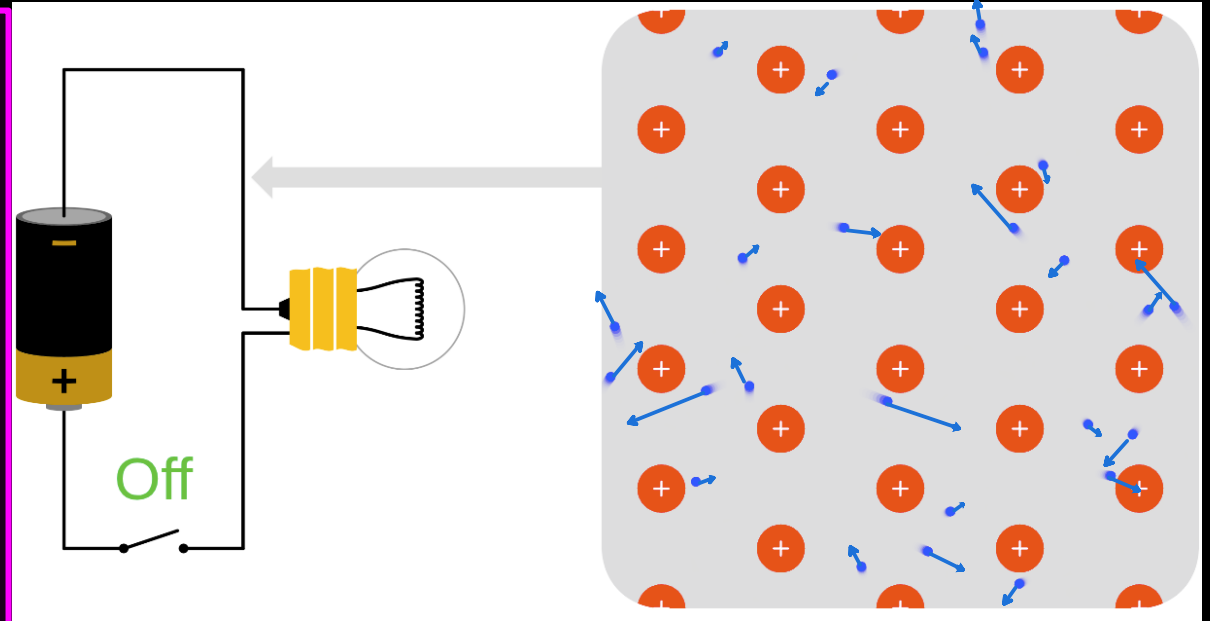


* In our discussions, we will focus only on solid conductors so that the current is carried by the negatively charged electrons in the background of fixed positive ions.

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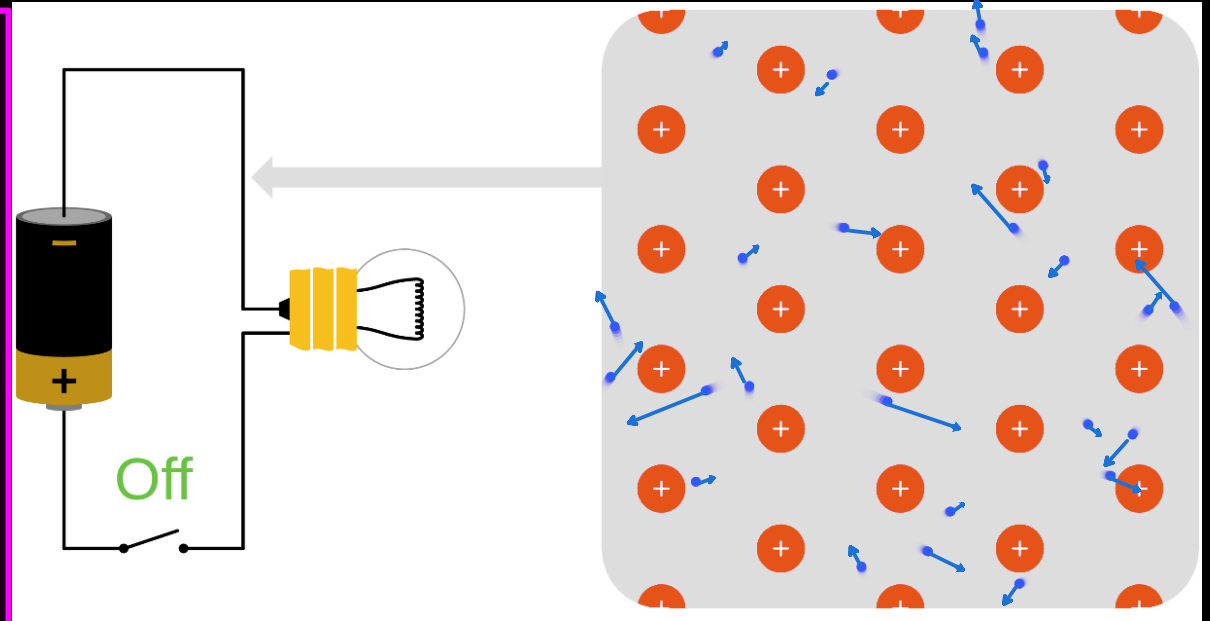
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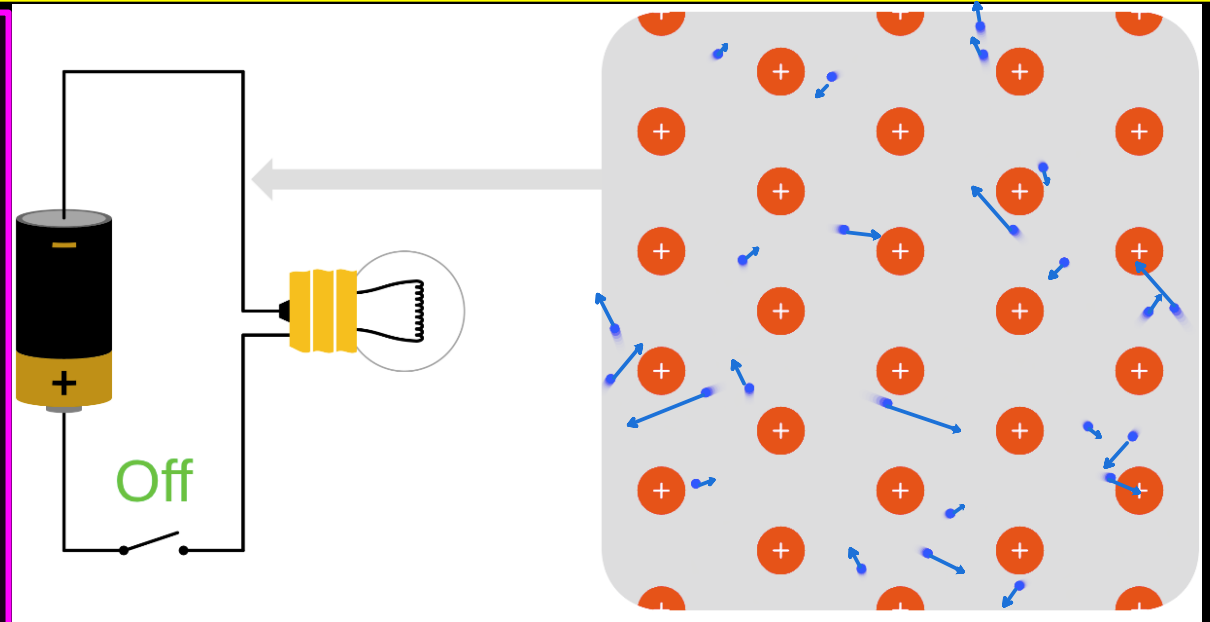
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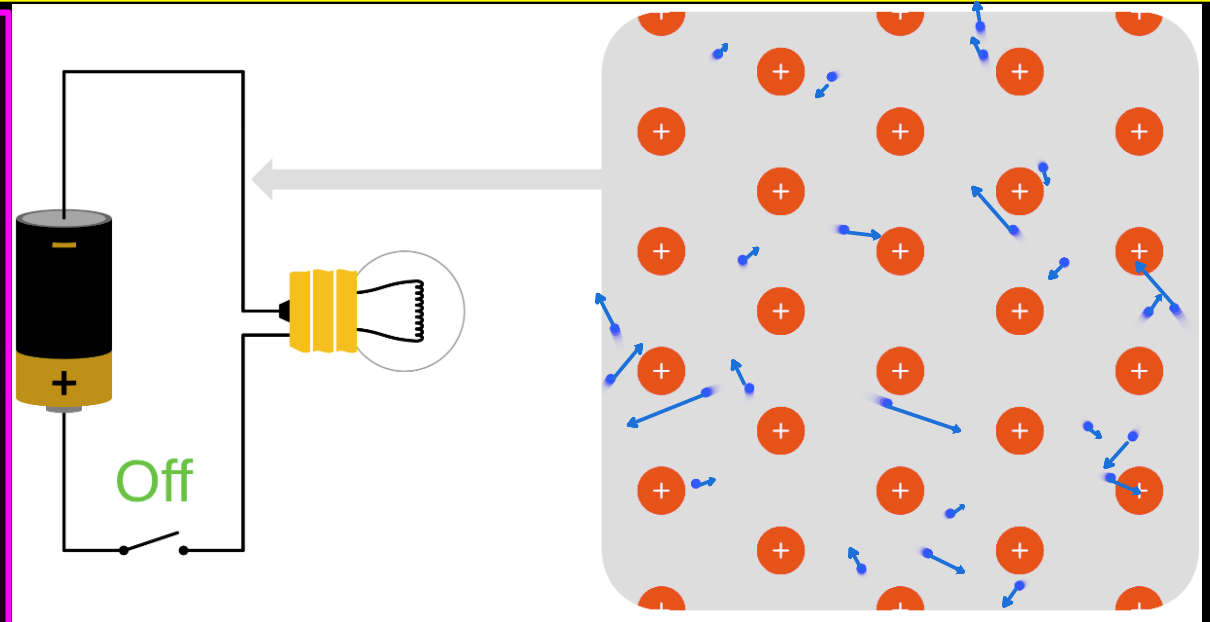


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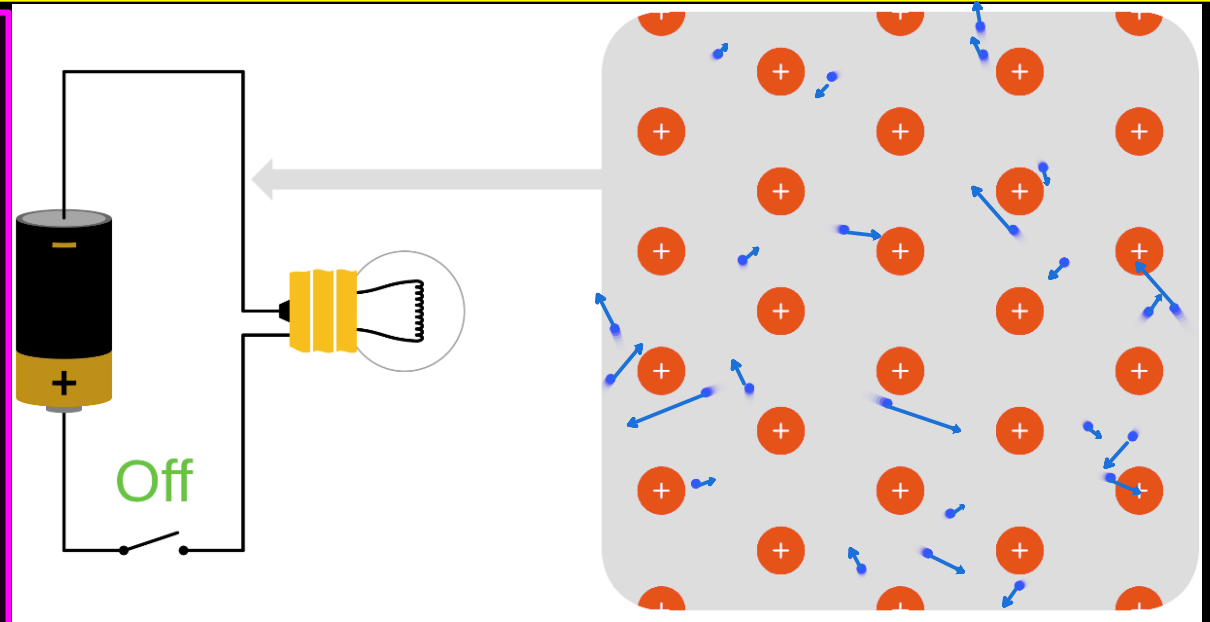


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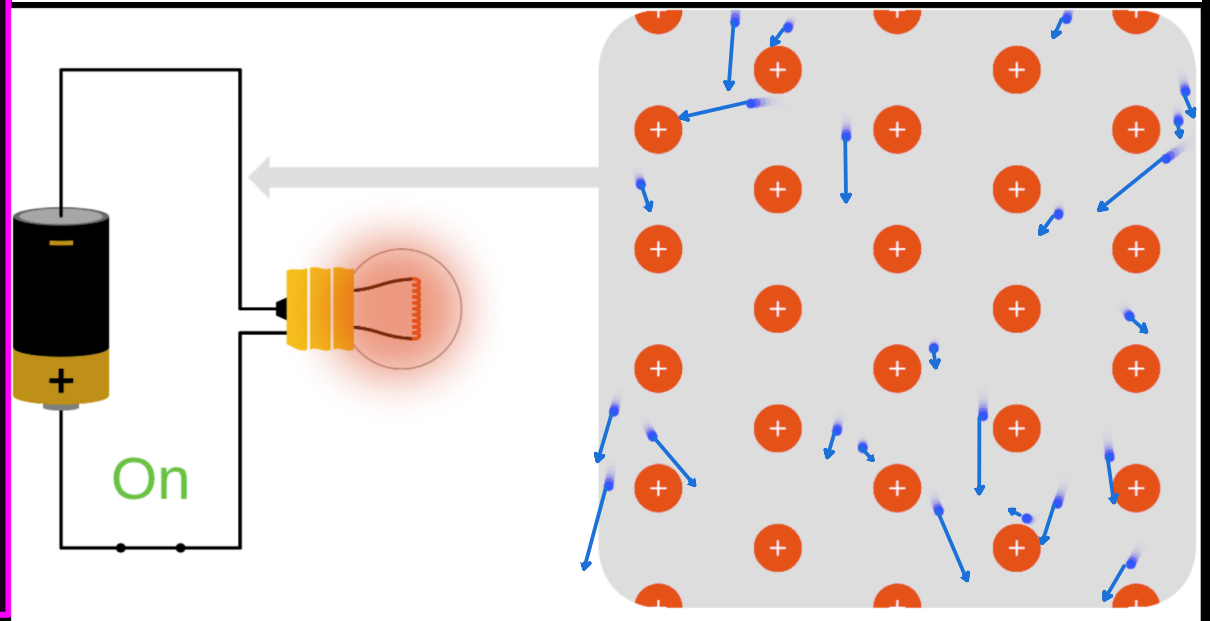
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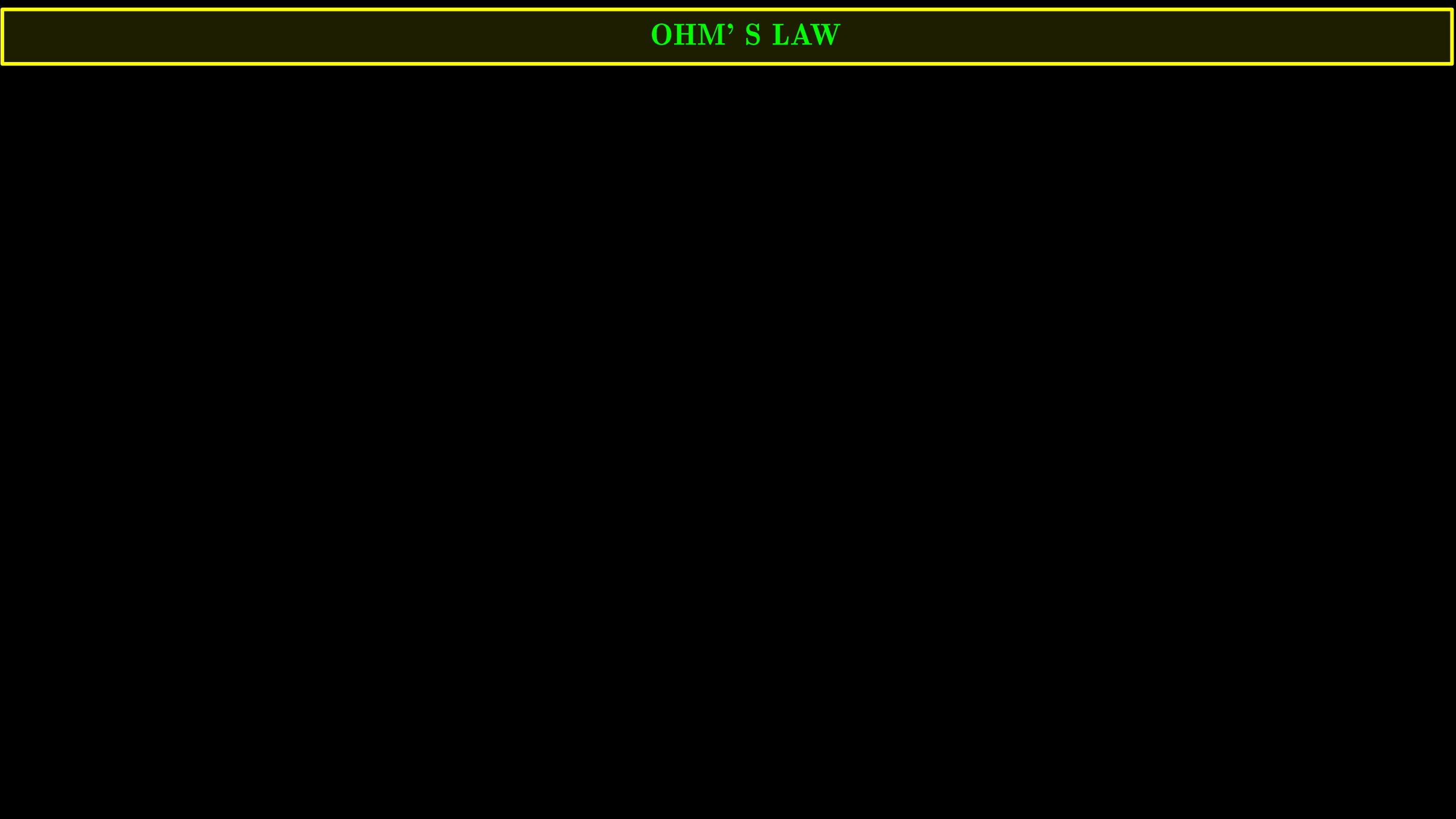


Case 2 : When electric field is present :

* The electrons will be accelerated due to this field, in the opposite direction of the field. The electrons, as long as they are moving, will constitute an electric current.



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$$R \propto \frac{l}{A} \text{ or } R = \rho \frac{l}{A}$$

* where the constant of proportionality called resistivity (ρ) depends on the material

CURRENT DENSITY AND EQUIVALENT FORM OF OHM'S LAW

Consider a rectangular block of material with length l , cross-sectional area A , and resistance R . A potential difference V is applied across the ends of the block, causing a current I to flow through it.

The current density J is defined as the current per unit area, and the electric field E is the potential difference per unit length.

Ohm's Law in terms of current density and electric field is given by:

$$J = \frac{I}{A} = \frac{V}{R A} = \frac{V}{l} \frac{1}{R A/l} = E \frac{1}{R A/l}$$

where $R A/l$ is the resistance of a unit cube of the material, known as the resistivity ρ .

Thus, the equivalent form of Ohm's Law is:

$$J = \frac{E}{\rho}$$

where ρ is the resistivity of the material.

The resistivity ρ is a property of the material and is independent of the dimensions of the block.

The conductivity σ is the reciprocal of the resistivity, $\sigma = 1/\rho$.

Thus, Ohm's Law can also be written as:

$$J = \sigma E$$

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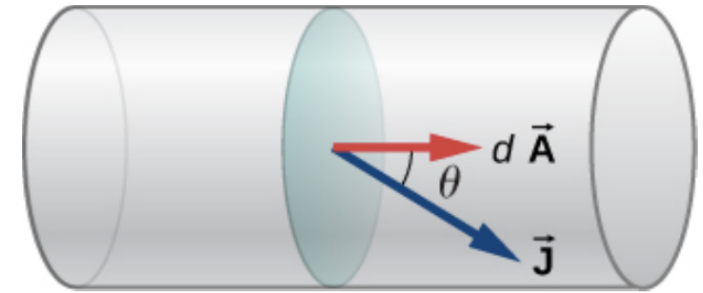


Figure 9.12 The current density $\vec{J}$ is defined as the current passing through an infinitesimal cross-sectional area divided by the area. The direction of the current density is the direction of the net flow of positive charges and the magnitude is equal to the current divided by the infinitesimal area.

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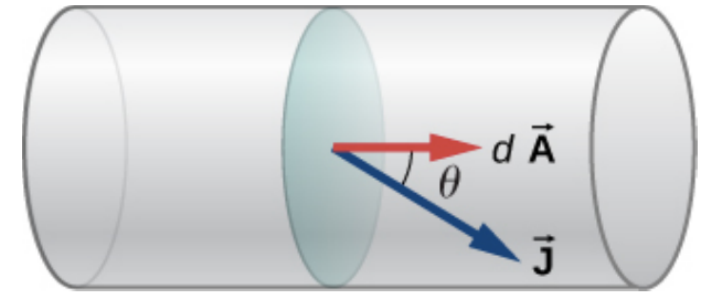


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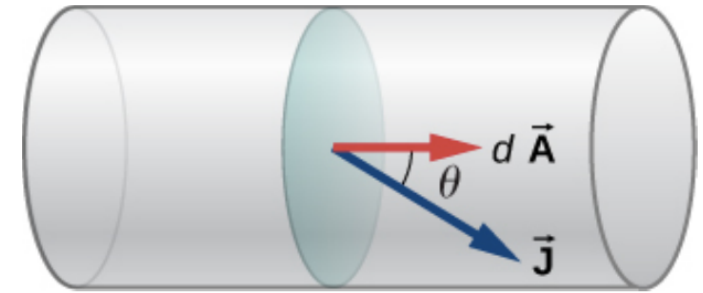


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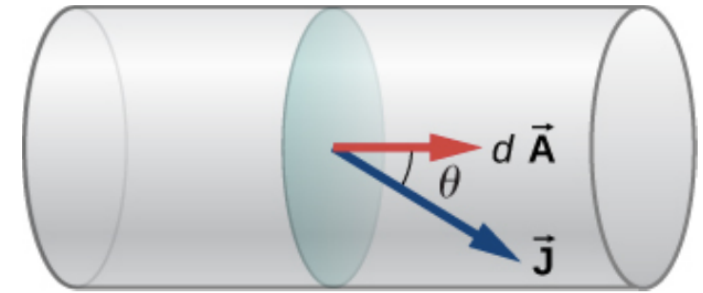


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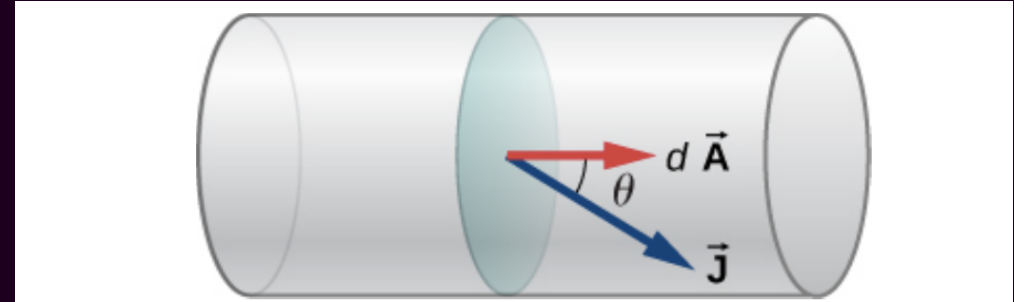


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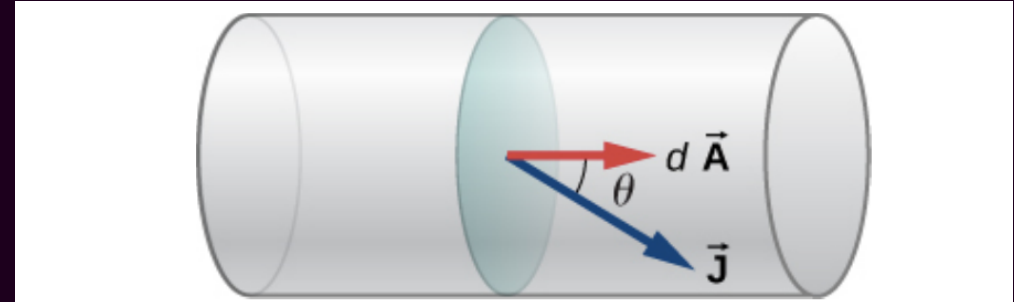


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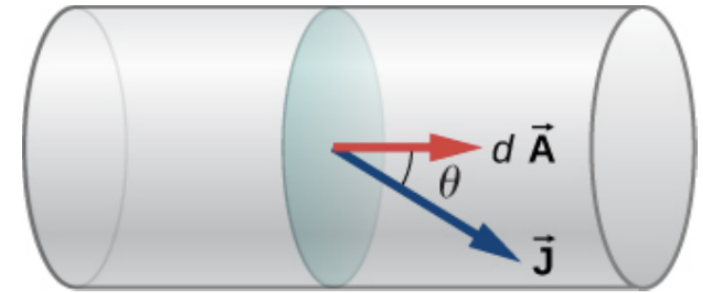


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3.6 A negligibly small current is passed through a wire of length 15 m and uniform cross-section $6.0 \times 10^{-7} \text{ m}^2$, and its resistance is measured to be $5.0 \, \Omega$. What is the resistivity of the material at the temperature of the experiment?

Ex 6 : A wire 50 cm long and 1 mm² in cross-section carries a current of 4 A when connected to a 2 V battery. The resistivity of the wire is

a) $2 \times 10^{-7} \Omega m$ b) $5 \times 10^{-7} \Omega m$

c) $4 \times 10^{-6} \Omega m$ d) $1 \times 10^{-6} \Omega m$

Ex 7: A wire of length L and cross-section area A has resistance R . What will be the resistance of the wire if it is stretched to twice its original length? Assume that the density and resistivity of material do not change when the wire is stretched.

Ex 8 : A wire of length L is drawn such that its diameter is reduced to half of its original diameter. If the initial resistance of the wire is $10\ \Omega$, its new resistance would be

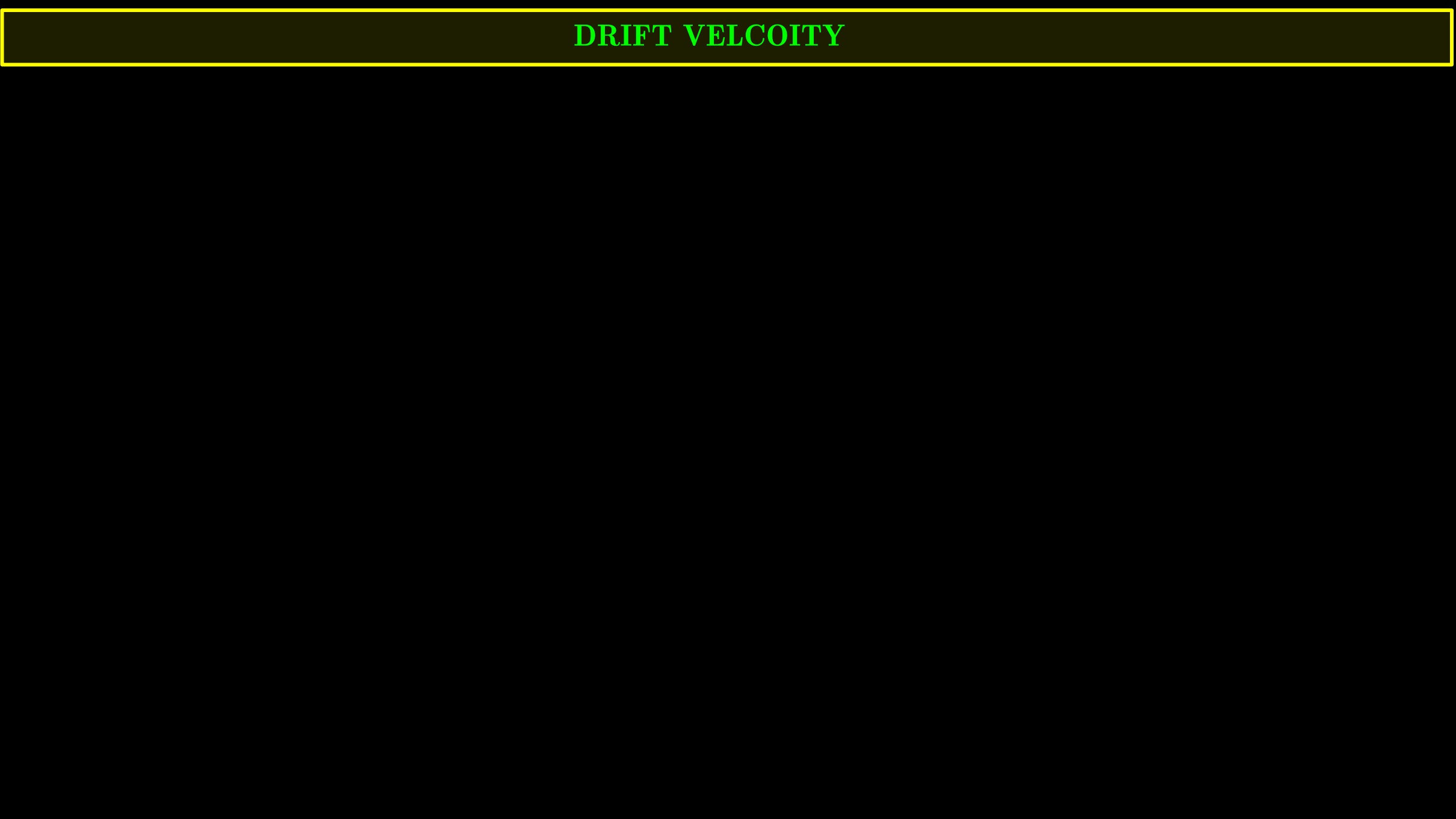
a) $40\ \Omega$

(b) $80\ \Omega$

(c) $120\ \Omega$

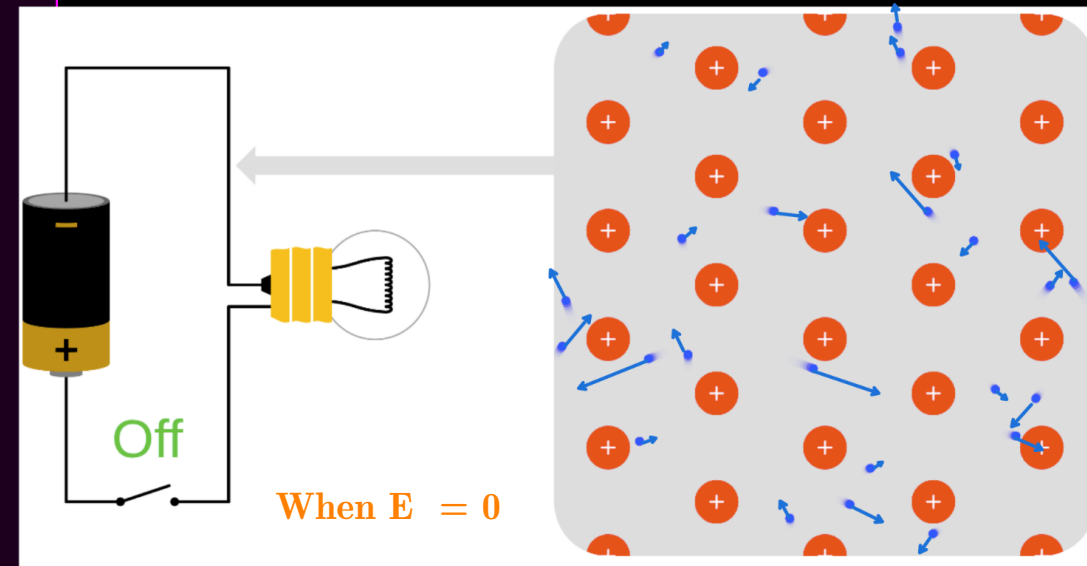
(d) $160\ \Omega$

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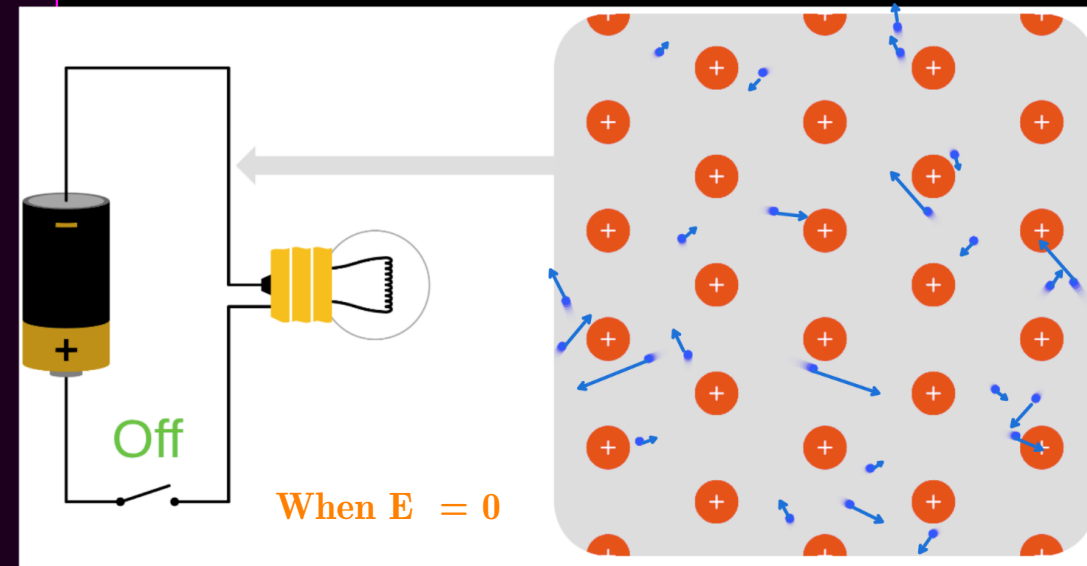
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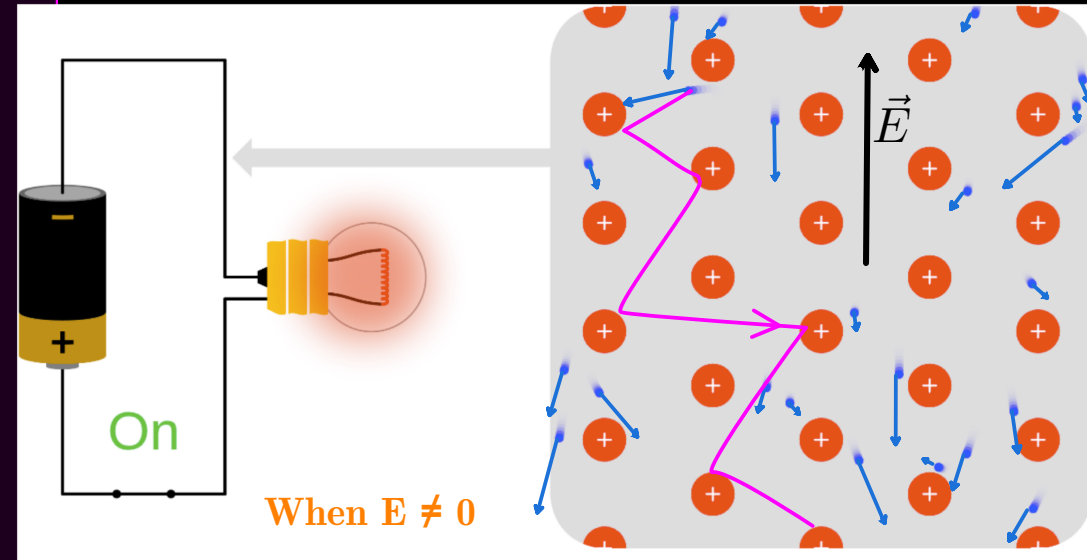
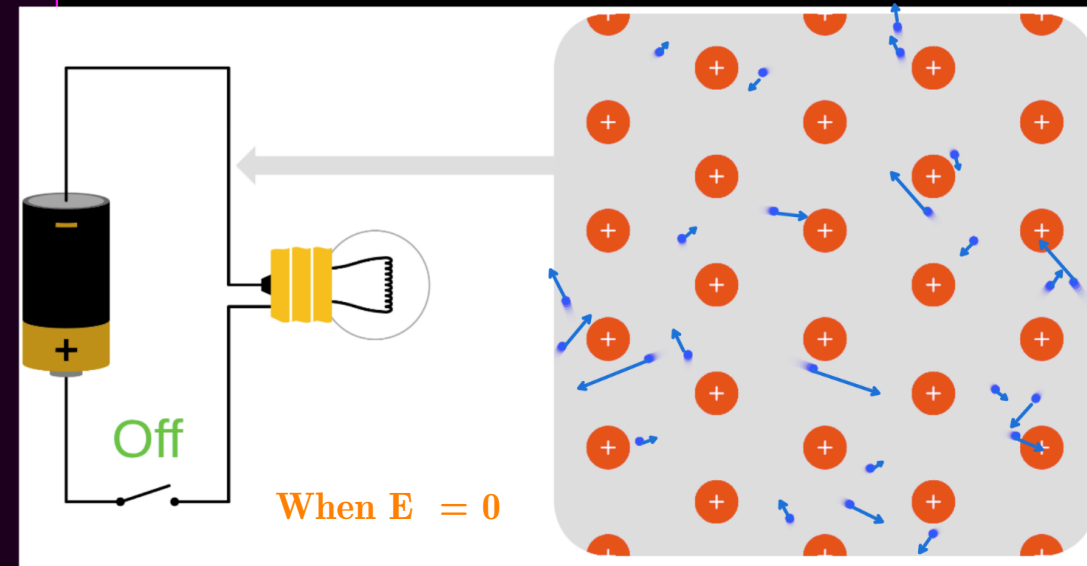
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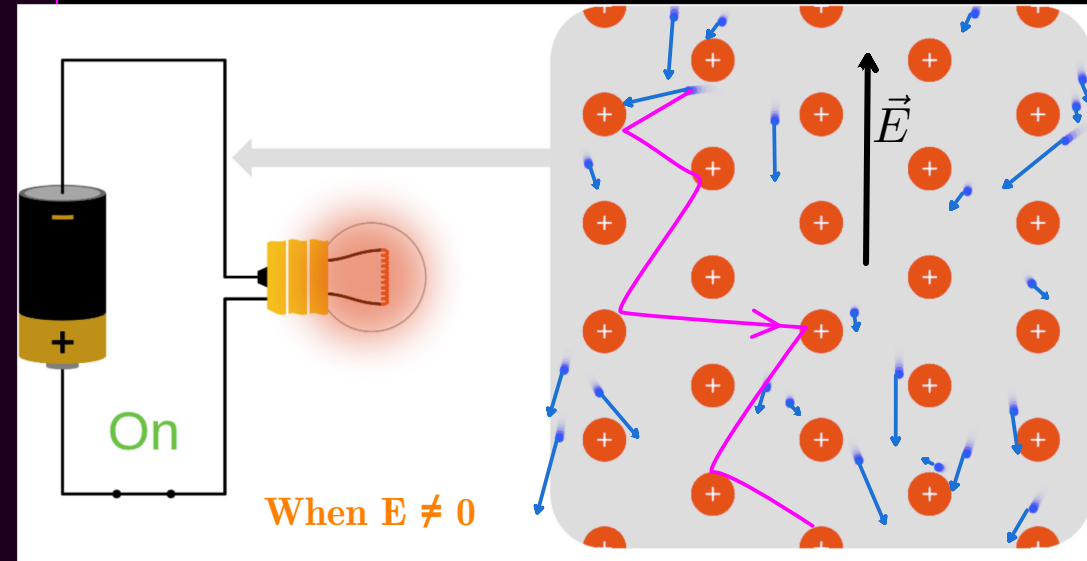
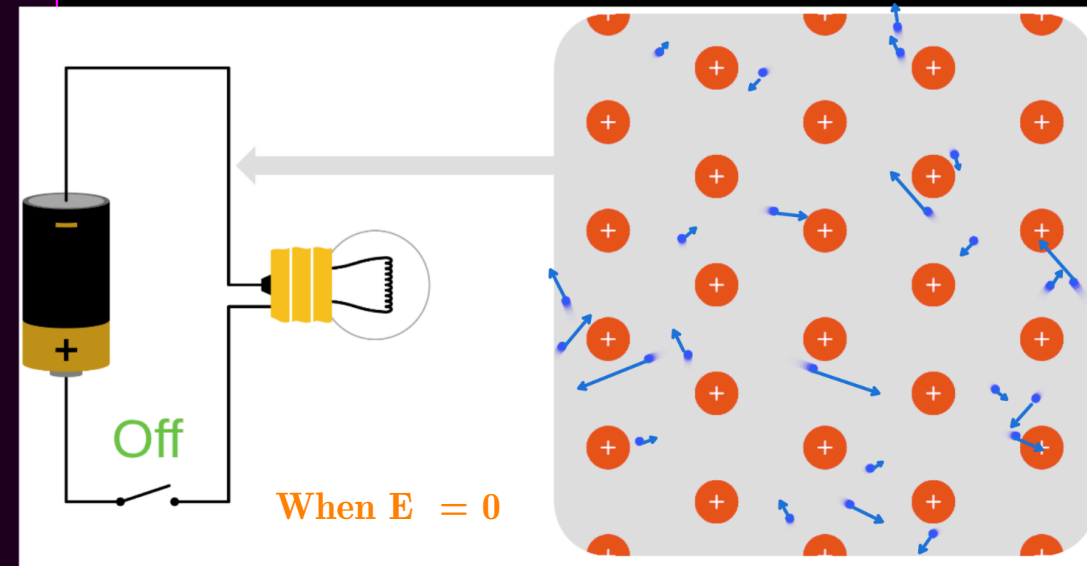
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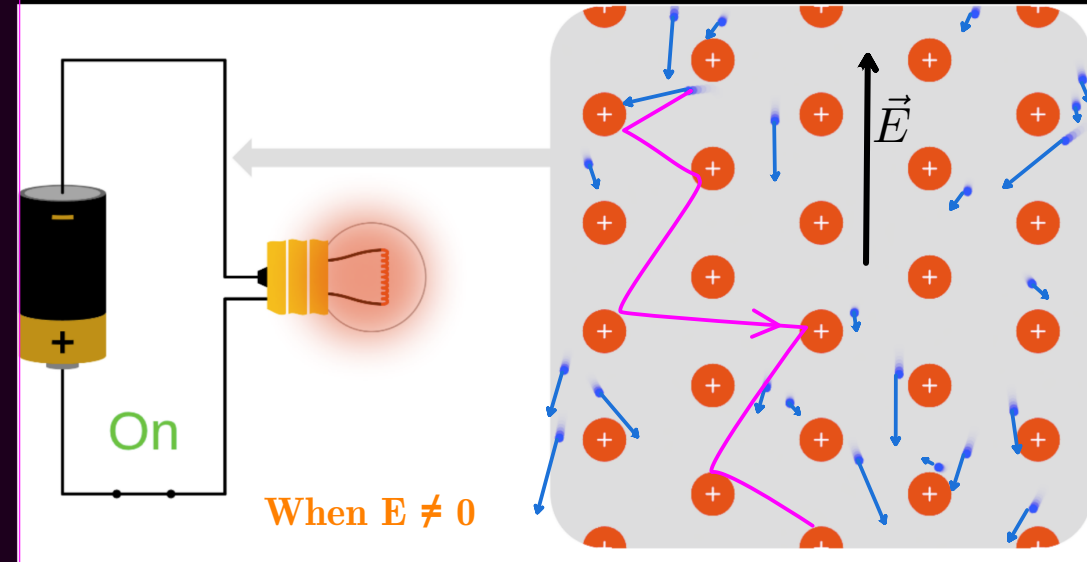
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$$\vec{F} = -e\vec{E} \quad \therefore \text{acceleration } \vec{a} = \frac{-e\vec{E}}{m_e}$$



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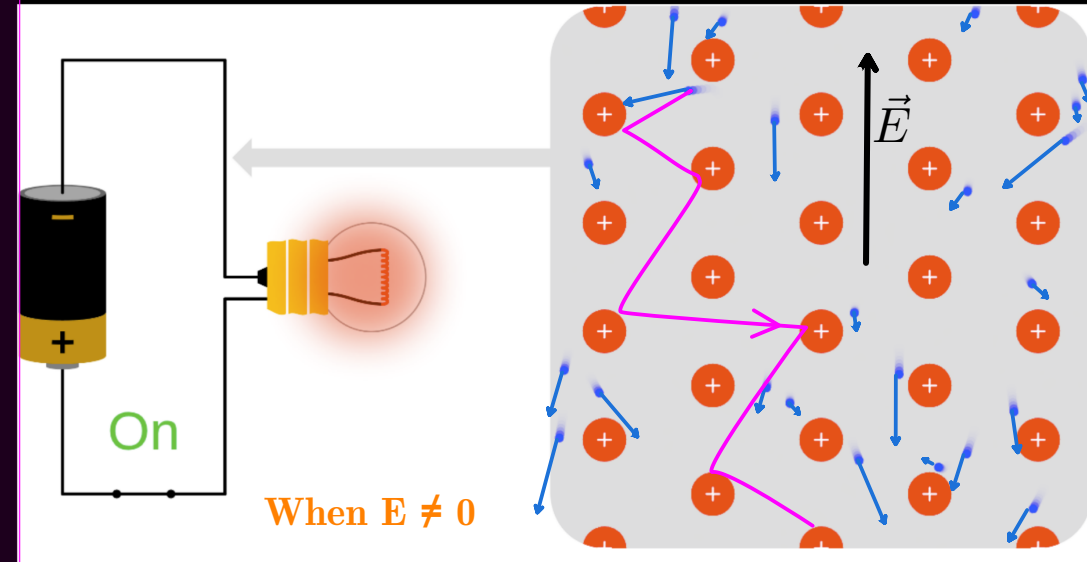
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- * The velocity gained by i^{th} electron between two successive collisions.
(t_i - time between two successive collision)

$$\vec{v}_i = \vec{u}_i + \vec{a} t_i$$

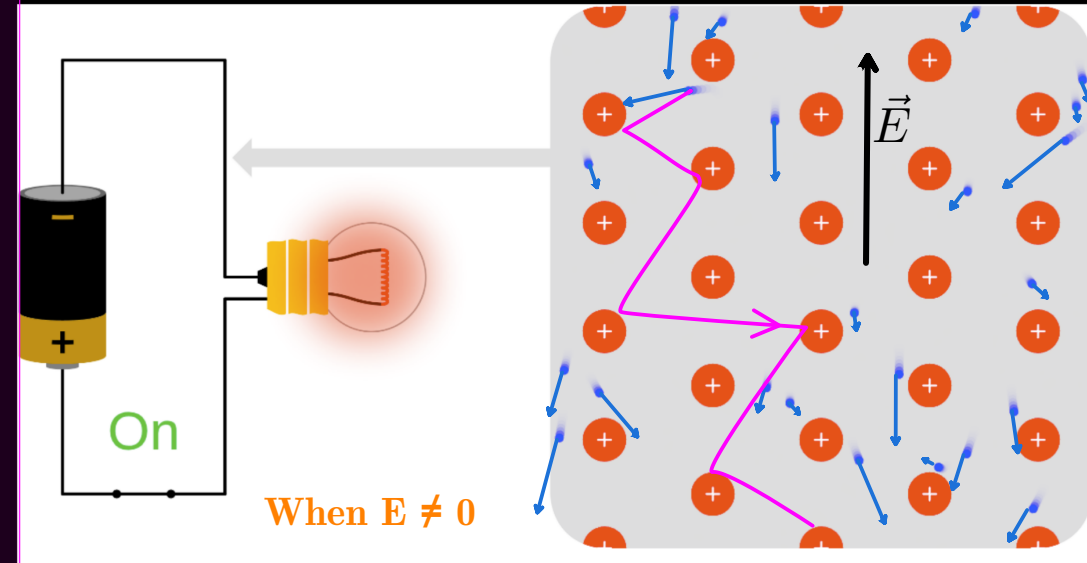


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- * The velocity gained by i^{th} electron between two successive collisions.

(t_i - time between two successive collision)

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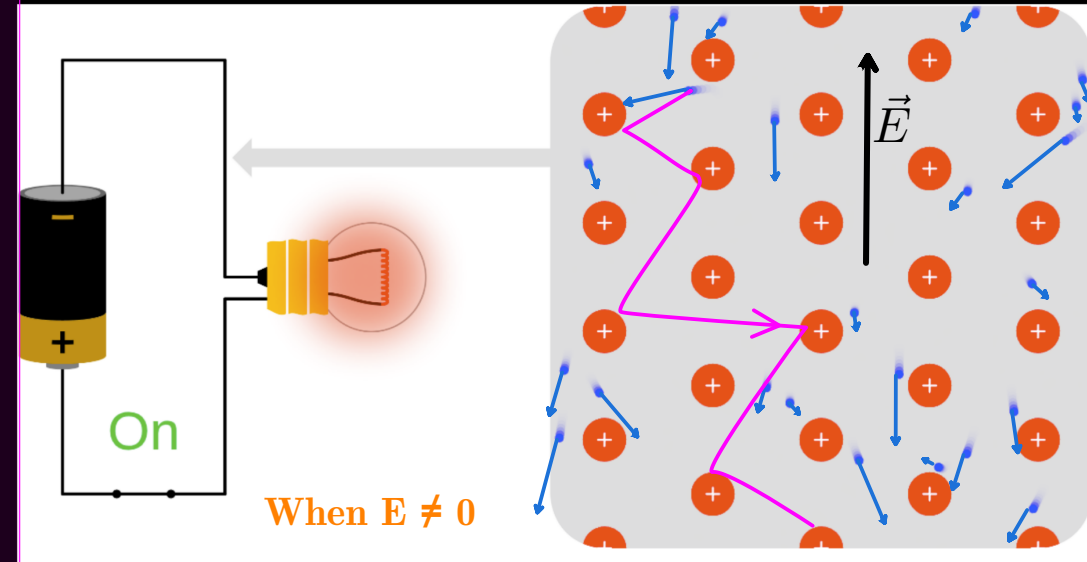
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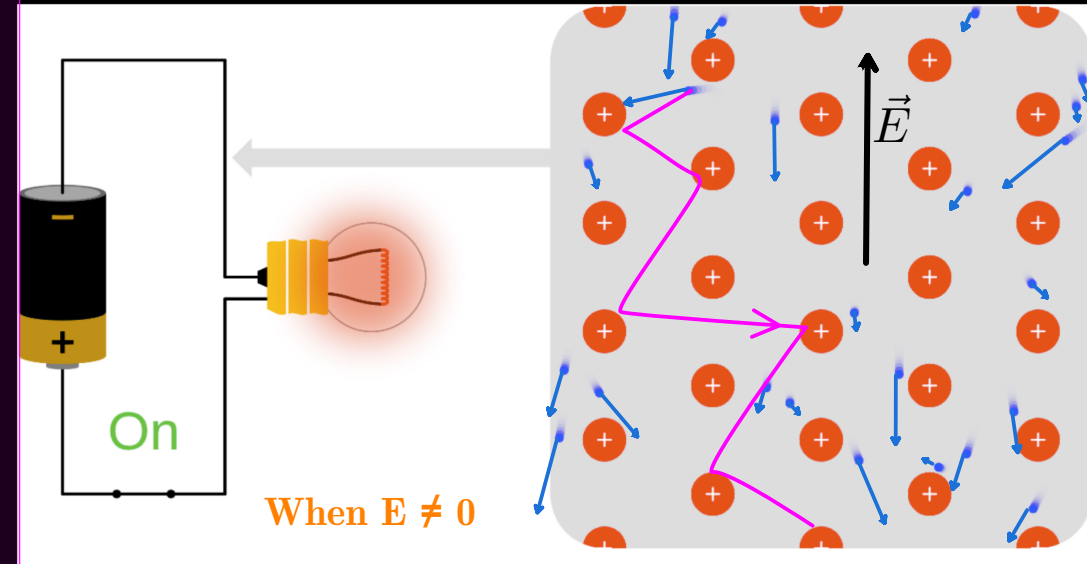
$$\vec{v}_{\text{avg.}} = \frac{1}{n} \sum_{i=1}^n v_i = \frac{1}{n} \sum_{i=1}^n u_i - \frac{e\vec{E}}{m_e} \sum_{i=1}^n \frac{t_i}{n}$$



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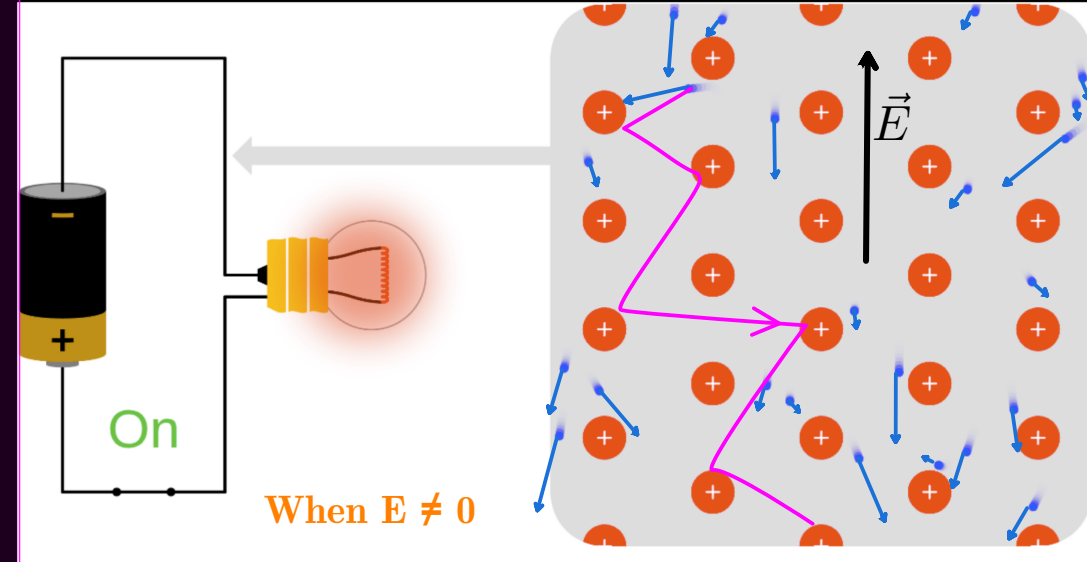
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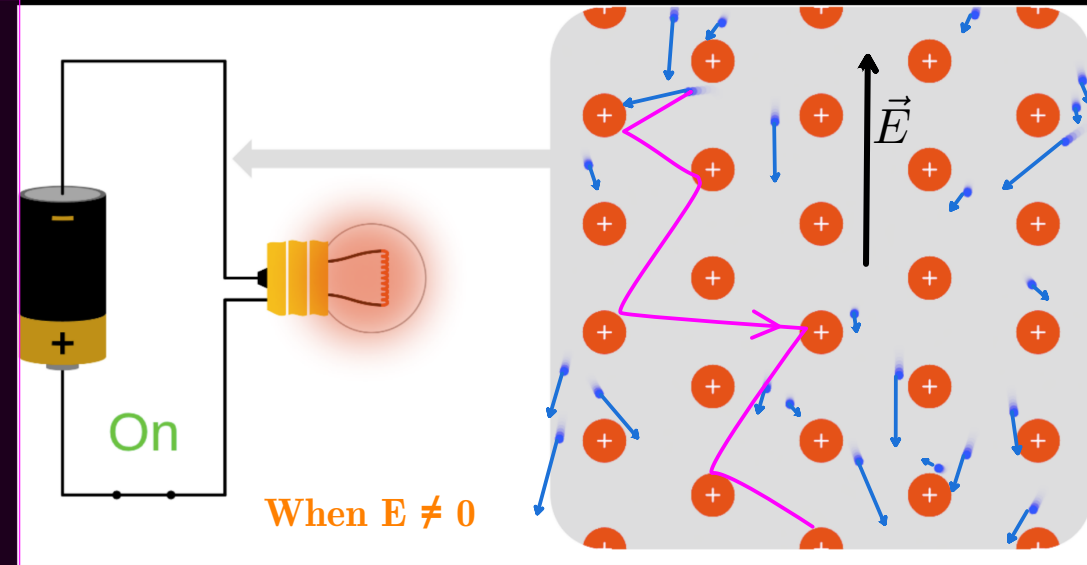
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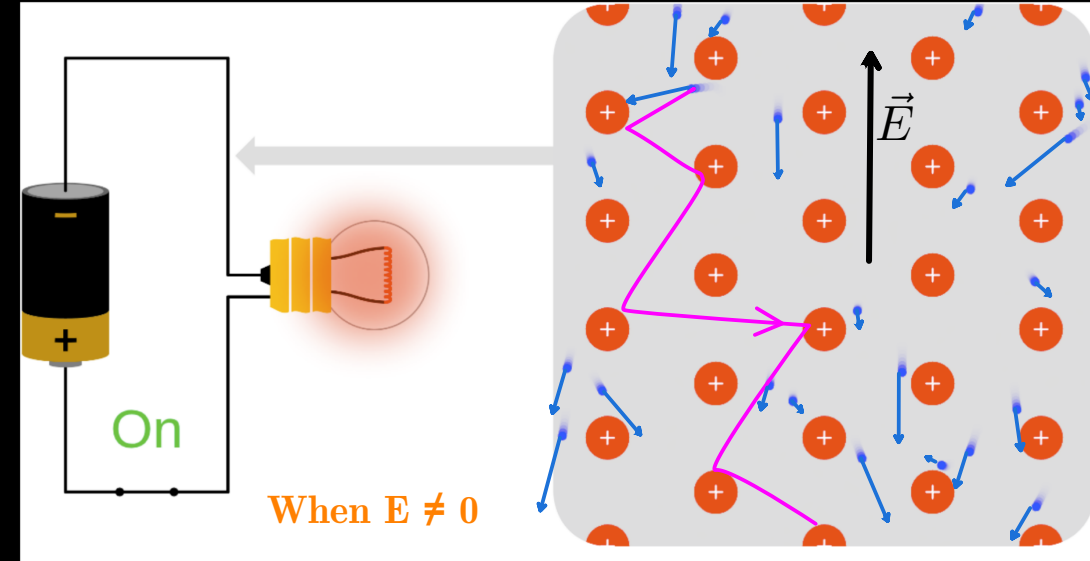
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- * **Drift velocity :** The average velocity with which the free electrons get drifted in the opposite direction of the applied electric field.

DRIFT VELOCITY

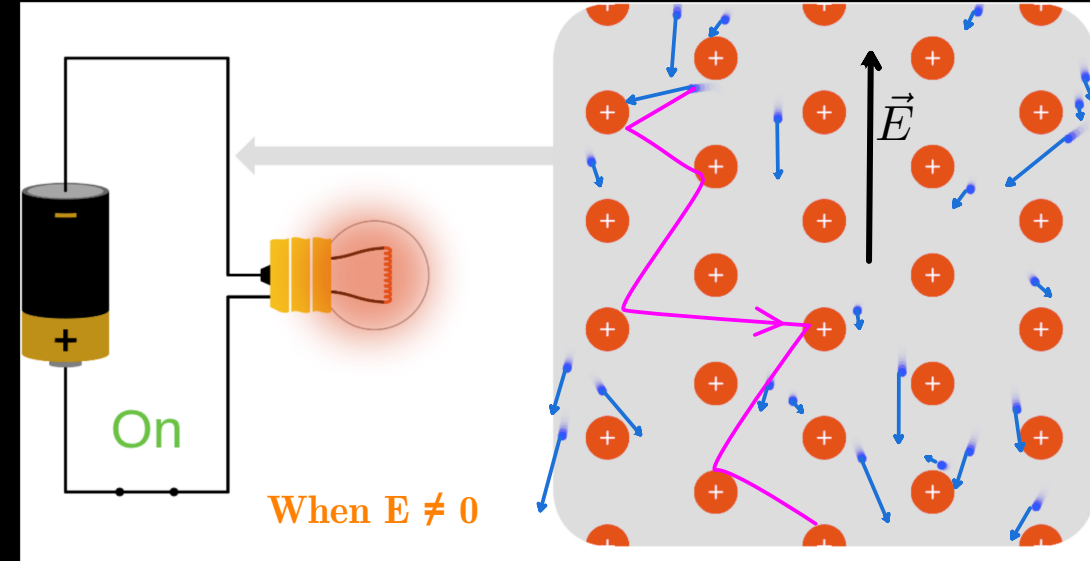
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* The drift velocity of electrons is very small in the order of 10^{-4} ms^{-1} .

